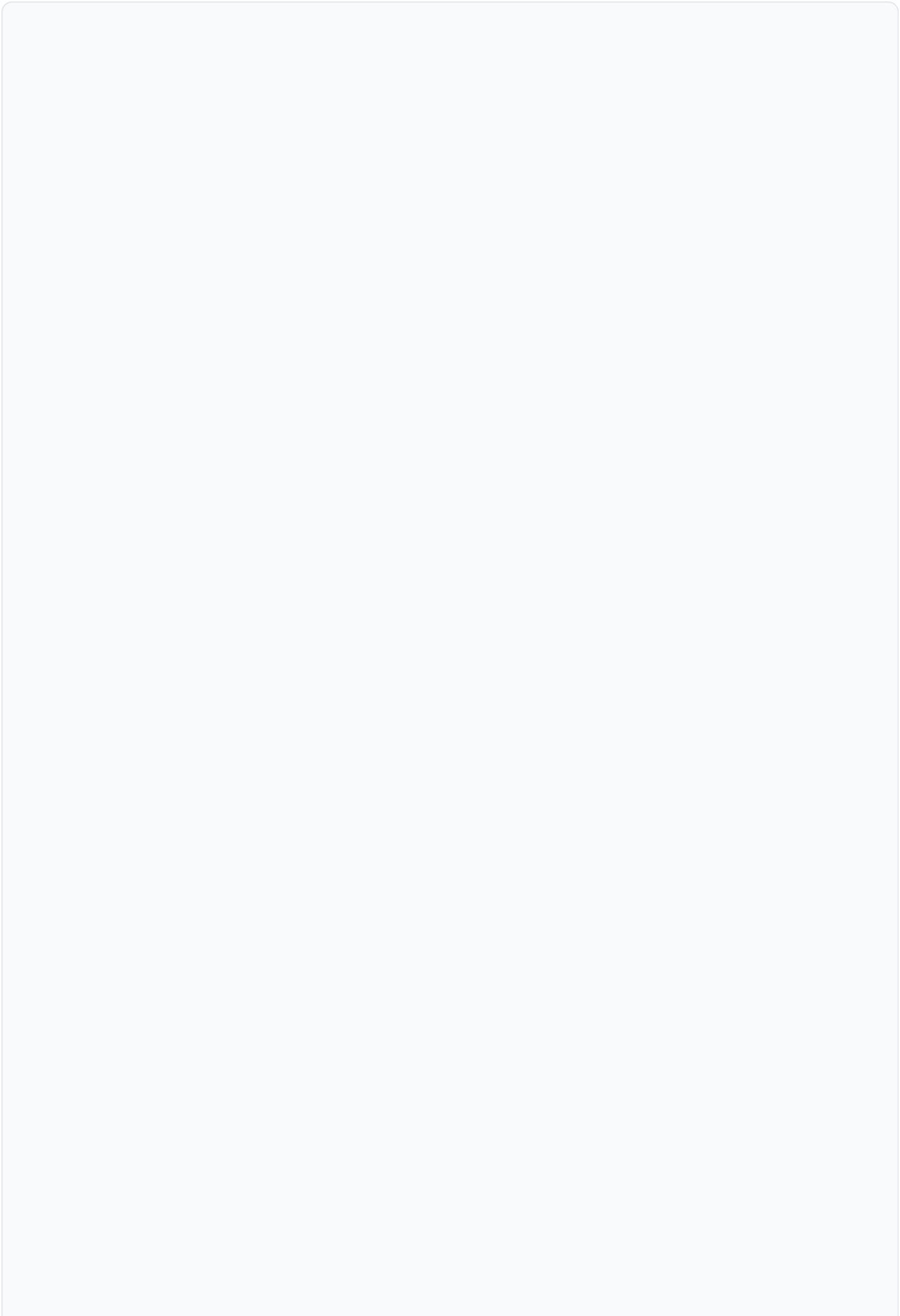


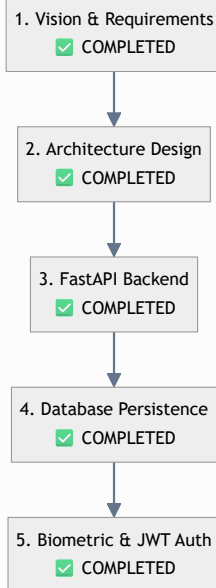
SecurePulse Mobile — Master Product & Engineering Roadmap

This document outlines the authoritative product roadmap, engineering milestones, architectural phases, and delivery commitments for **SecurePulse Mobile** (Enterprise Cybersecurity & SOC Mobile Operations Platform).

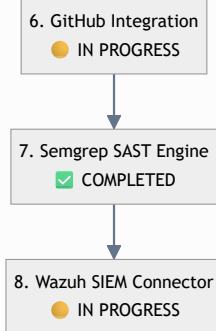
High-Level Product Architecture & Milestone Timeline



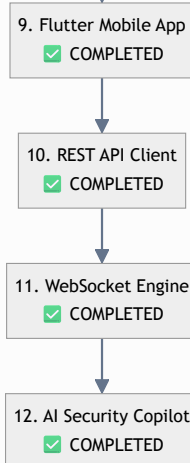
Phase 1 - 5: Foundation & Security Core



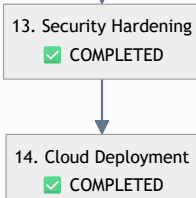
Phase 6 - 8: Toolchain Integrations

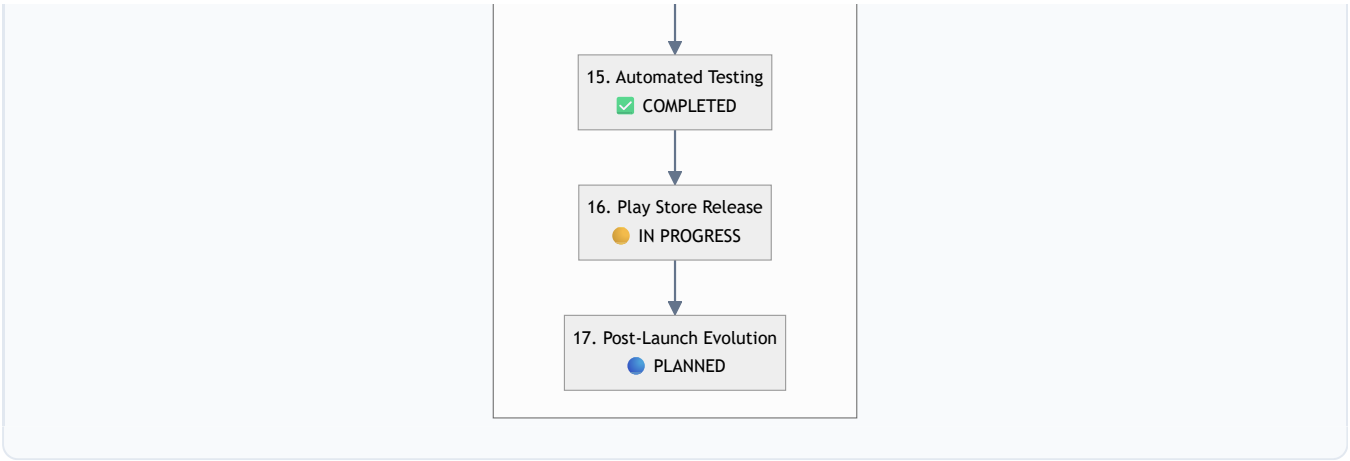


Phase 9 - 12: Mobile & Real-Time Engine



Phase 13 - 17: Production Release & Evolution





📌 Comprehensive 17-Phase Product Roadmap

1. Vision & Requirements

- **Objective:** Define problem domain, user personas, operational constraints, and multi-tier cybersecurity monitoring requirements.
- **Tasks:**
 - Establish user stories for Tier-1/2/3 SOC Analysts, Incident Responders, and Security Leadership.
 - Define core latency benchmarks (< 500ms API response, < 50ms real-time event streaming).
 - Formulate air-gapped zero-latency Demo Mode specification for executive presentations.
- **Technology:** Markdown Specifications, Agile User Story Mapping.
- **Deliverables:** Project Charter, Functional Requirements Matrix, System Specification Dossier.
- **Validation:** Stakeholder review against real-world SOC operational friction points.
- **Status:** ✓ COMPLETED

2. Architecture Design

- **Objective:** Formulate the multi-tier Clean Architecture, reactive state management model, and containerized cloud deployment topology.
- **Tasks:**
 - Design 4-layer Feature-First Clean Architecture (Presentation, State, Domain, Data).
 - Establish Riverpod 2.5.1 provider dependencies and GoRouter 5-tab shell navigation.
 - Design dual-mode repository dispatching architecture (Demo Mode vs. Live Network Mode).
- **Technology:** Flutter/Dart, Mermaid Diagrams, Clean Architecture Design Patterns.
- **Deliverables:** docs/MINDMAP.md , Architecture Specification in docs/PROJECT_REPORT.md .
- **Validation:** Architectural design review verifying decoupling of UI widgets from data layers.
- **Status:** ✓ COMPLETED

3. FastAPI Backend

- **Objective:** Develop an asynchronous, high-performance REST and WebSocket API microservice.
 - **Tasks:**
 - Implement route handlers (`/health` , `/v1/auth` , `/v1/dashboard` , `/v1/repositories` , `/v1/soc/alerts` , `/v1/ai` , `/v1/reports`).
 - Enforce Pydantic schema validation for all request and response envelopes.
 - Configure CORS middleware, timeout gates, and ASGI worker concurrency.
 - **Technology:** Python 3.11, FastAPI, Uvicorn, Pydantic, CORS Middleware.
 - **Deliverables:** Deployed FastAPI microservice on Render Cloud (`https://secureguard-backend-7eqm.onrender.com`).
 - **Validation:** Automated `GET /health` health probes and REST endpoint status verification.
 - **Status:**  COMPLETED
-

4. Database Persistence

- **Objective:** Implement cloud relational storage and client-side secure offline caching.
 - **Tasks:**
 - Provision managed PostgreSQL database with SSL connection pooling on cloud infrastructure.
 - Implement local key-value offline storage box (`HiveStorageService` , `securepulse_cache`).
 - Configure local cache synchronization for scan findings and executive posture scores.
 - **Technology:** PostgreSQL 16 (Cloud SSL), Hive 2.2.3 (Mobile Client).
 - **Deliverables:** `lib/core/storage/hive_storage_service.dart` , Cloud PostgreSQL database schemas.
 - **Validation:** Cold boot offline data recovery test validating instant cache hydration.
 - **Status:**  COMPLETED
-

5. Authentication & Credential Security

- **Objective:** Implement hardware-backed biometric authentication and stateless JWT session management.
 - **Tasks:**
 - Implement `local_auth` biometric challenge gate (Face ID / Android BiometricPrompt).
 - Integrate `FlutterSecureStorage` utilizing Android Keystore (AES-256 / RSA) and Apple Keychain.
 - Build `AuthRepositoryImpl` with login, logout, and token auto-injection interceptors.
 - **Technology:** `local_auth 2.3.0` , `flutter_secure_storage 9.2.4` , JWT (RFC 7519), bcrypt.
 - **Deliverables:** `lib/features/auth/` , `lib/core/services/biometric_service.dart` , `lib/core/storage/secure_storage_service.dart` .
 - **Validation:** Keystore read/write verification and biometric session recovery tests (`TC-06`).
 - **Status:**  COMPLETED
-

6. GitHub Integration

- **Objective:** Synchronize enterprise codebase telemetry, branch metadata, and security audit status.
 - **Tasks:**
 - Implement `RepositoryModel` tracking language, branch, health grades (A-F), and vulnerability counts.
 - Build UI for monitored repositories list and repository detail views.
 - [In Progress] Integrate interactive in-app OAuth2 GitHub App consent deep-linking loop.
 - **Technology:** GitHub REST API v3, Webhooks, GoRouter.
 - **Deliverables:** `lib/features/repositories/`, `RepositoryRepositoryImpl`.
 - **Validation:** Repository inventory rendering and scan trigger dispatch (`TC-08`).
 - **Status:** 🟡 IN PROGRESS (Direct token session functional; OAuth2 deep-linking in roadmap)
-

7. Semgrep SAST Engine

- **Objective:** Ingest and categorize static application security testing findings and trigger on-demand scans.
 - **Tasks:**
 - Implement `ScanModel` and `FindingModel` with line numbers, file paths, and CWE tags.
 - Build scan detail screens with severity distribution badges (Critical, High, Medium, Low).
 - Wire single-tap "Trigger Scan" action dispatching `POST /v1/repositories/{id}/scan`.
 - **Technology:** Semgrep SAST JSON Schemas, Riverpod Providers.
 - **Deliverables:** `lib/features/scans/`, `lib/features/findings/`, `ScanRepository`, `FindingRepository`.
 - **Validation:** Verification of finding counts, CWE tags, and scan status state transitions.
 - **Status:** ✅ COMPLETED
-

8. Wazuh SIEM Integration

- **Objective:** Ingest host intrusion detection logs, brute-force alarms, and system integrity events.
 - **Tasks:**
 - Model SIEM alerts (`AlertModel`) capturing syslog payloads, attacker IP, and remediation advice.
 - Build Alert Detail screen with raw syslog inspection and "Quarantine IP" action button.
 - [In Progress] Direct bidirectional integration with Wazuh Manager API (`port 55000`) for remote daemon management.
 - **Technology:** Wazuh SIEM Syslog Parser, REST API, WebSockets.
 - **Deliverables:** `lib/features/alerts/`, `AlertsRepositoryImpl`.
 - **Validation:** Alert filtering by severity (Critical, High, Medium, Low) and status mutation (`TC-09`).
 - **Status:** 🟡 IN PROGRESS (Syslog feed & status triage functional; Manager API in roadmap)
-

9. Flutter Mobile Application

- **Objective:** Build a responsive, cyber-themed cross-platform mobile user interface.
 - **Tasks:**
 - Implement Material 3 Cyber Dark Obsidian (#0A0E1A) and Clean Light themes in `AppTheme` .
 - Build 5-tab shell navigation with animated navbar transitions (`SGNavbar`).
 - Implement executive dashboard with interactive posture gauge and `fl_chart` donut breakdown.
 - **Technology:** Flutter 3.x, Dart 3.x, GoRouter 13.2.0, `fl_chart` 0.66.2, Google Fonts.
 - **Deliverables:** Complete `lib/` codebase, `SecurePulseApp` root widget.
 - **Validation:** Widget pump testing (`TC-15`) and 60.0–120.0 fps frame rate profiling.
 - **Status:**  COMPLETED
-

10. REST API Client & Interceptors

- **Objective:** Implement a resilient, typed HTTP network client with error classification.
 - **Tasks:**
 - Configure Dio 5.4.1 client with dynamic Bearer token headers and 12-second timeout gates.
 - Implement `ApiException` translation mapping 401, 403, 404, 500, and socket timeouts.
 - Add dynamic runtime `updateBaseUrl()` support for seamless environment switching.
 - **Technology:** Dio 5.4.1, Typed Dart Exceptions.
 - **Deliverables:** `lib/core/network/api_client.dart` , `lib/core/error/api_exception.dart` .
 - **Validation:** Unit tests verifying header injection, token removal, and base URL updates (`TC-01` – `TC-05`).
 - **Status:**  COMPLETED
-

11. WebSocket Real-Time Engine

- **Objective:** Deliver sub-second SIEM threat streaming with automatic reconnection resilience.
 - **Tasks:**
 - Build `WebSocketService` with dynamic URL scheme conversion (`http→ws` , `https→wss`).
 - Broadcast incoming threat events to reactive Riverpod stream provider (`websocketAlertStreamProvider`).
 - Implement exponential backoff reconnect loops and HTTP long-polling fallback.
 - **Technology:** Dart `web_socket_channel` 2.4.5 , StreamControllers, Riverpod Streams.
 - **Deliverables:** `lib/core/network/websocket_service.dart` .
 - **Validation:** Live scheme translation and Demo mode disconnection isolation tests (`TC-12` – `TC-14`).
 - **Status:**  COMPLETED
-

12. AI Security Copilot

- **Objective:** Provide conversational cybersecurity remediation advice and actionable code patches.

- **Tasks:**
 - Build interactive streaming chat UI with markdown rendering and syntax-highlighted code cards.
 - Implement offline heuristic engine generating remediation patches for CVE-2024-3094, SQLi, and AWS secret leaks.
 - Connect live cloud LLM bridge via `POST /v1/ai/chat` for dynamic conversational analysis.
 - **Technology:** Flutter Markdown, Heuristic Rule Matchers, Cloud LLM Proxy.
 - **Deliverables:** `lib/features/ai/`, `AiRepositoryImpl`.
 - **Validation:** Heuristic code generation tests (`TC-10`) and interactive chat prompt rendering.
 - **Status:**  COMPLETED
-

13. Security Hardening & Cryptographic Audit

- **Objective:** Protect client perimeter, vault credentials, and guarantee report non-repudiation.
 - **Tasks:**
 - Implement STRIDE threat mitigations across all 6 attack categories.
 - Build `ReportPdfService` generating vector compliance PDFs (SOC 2, ISO 27001, PCI-DSS).
 - Embed on-device SHA256 cryptographic audit digest stamped onto all generated PDF pages.
 - **Technology:** `pdf 3.12.0`, `crypto 3.0.3` (SHA256), Android Keystore, iOS Keychain.
 - **Deliverables:** `lib/core/services/report_pdf_service.dart`, Security Design in `docs/PROJECT_REPORT.md`.
 - **Validation:** Cryptographic audit digest verification and zero-leakage logging audit.
 - **Status:**  COMPLETED
-

14. Cloud Deployment & Containerization

- **Objective:** Automate cloud microservice deployment and provide containerized web distribution.
 - **Tasks:**
 - Create multi-stage `Dockerfile` compiling Flutter Web and serving via `nginx:alpine` on port 80.
 - Configure Render Cloud Web Service with Git auto-deploy webhooks and SSL/TLS termination.
 - Implement automated `/health` probes ensuring 99.9% service uptime.
 - **Technology:** Docker, Nginx, Render Cloud CI/CD, Let's Encrypt TLS 1.3.
 - **Deliverables:** `Dockerfile`, Render Web Service (`secureguard-backend-7eqm.onrender.com`).
 - **Validation:** Cloud health probe curl verification and Docker web container execution.
 - **Status:**  COMPLETED
-

15. Automated Testing & Verification

- **Objective:** Ensure 100% test pass rate across core business logic, API contracts, and widget trees.
- **Tasks:**
- Develop comprehensive test suite in `test/api_integration_test.dart` and `test/widget_test.dart`.

- Execute static analysis audit via `flutter analyze` with zero warnings.
 - Benchmark runtime performance (frame rates, memory consumption, latency).
 - **Technology:** `flutter_test`, `flutter_lints 3.0.2`.
 - **Deliverables:** 15/15 passing automated test suite, `flutter analyze` clean report.
 - **Validation:** **15 / 15 Tests Passed (100% Pass Rate)**, `flutter analyze` (0 issues).
 - **Status:**  COMPLETED
-

16. Google Play Store Release Preparation

- **Objective:** Package, sign, and prepare the production Android release artifact for commercial distribution.
 - **Tasks:**
 - Compile optimized standalone release APK (`securepulse-release.apk` , **63.6 MB**) via `flutter build apk --release`.
 - [In Progress] Generate production JKS keystore and configure `key.properties` for App Bundle (`AAB`) signing.
 - Prepare Google Play Console Data Safety declaration and privacy policy documentation.
 - **Technology:** Android Gradle Plugin, ProGuard / R8, Google Play Console.
 - **Deliverables:** `securepulse-release.apk` , Play Store publishing assets.
 - **Validation:** Successful sideload installation and biometric execution on physical Android 14 device.
 - **Status:**  IN PROGRESS (Release APK compiled; Play Store console submission pending)
-

17. Post-Launch Evolution & Advanced SOAR

- **Objective:** Continuously evolve SecurePulse into an autonomous mobile security operations ecosystem.
 - **Tasks:**
 - Implement autonomous AI remediation generating direct GitHub Pull Requests.
 - Build multi-cloud SOAR connectors (AWS Network ACLs, Cloudflare WAF, Cortex XSOAR).
 - Introduce multi-tenancy organization switcher for Managed Security Service Providers (MSSPs).
 - Integrate on-device quantized SLM (Gemma-2B / Llama-3-1B) via ONNX Runtime.
 - **Technology:** ONNX Runtime, AWS SDK, Cloudflare API, Redis Pub/Sub.
 - **Deliverables:** Enterprise SOAR Playbook connectors, Multi-Tenant Portal.
 - **Validation:** Automated perimeter block simulation and multi-tenant isolation benchmarks.
 - **Status:**  PLANNED
-

Summary of Phase Status

Category	Total Phases	Completed ()	In Progress ()	Planned ()	Not Implemented ()
Foundation & Architecture (Phases 1–5)	5	5	0	0	0
Toolchain Integrations (Phases 6–8)	3	1	2	0	0
Mobile & Real-Time Engine (Phases 9–12)	4	4	0	0	0
Production & Release (Phases 13–17)	5	3	1	1	0
OVERALL TOTAL	17	13 (76.5%)	3 (17.6%)	1 (5.9%)	0 (0.0%)